

Household Energy Consumption Analysis – Executive Summary

The city commissioned this analysis to better understand household energy usage patterns and design effective subsidy schemes for energy efficiency. This summary presents the key findings from the dataset of household electricity and gas usage, as well as targeted recommendations for energy-saving programs.

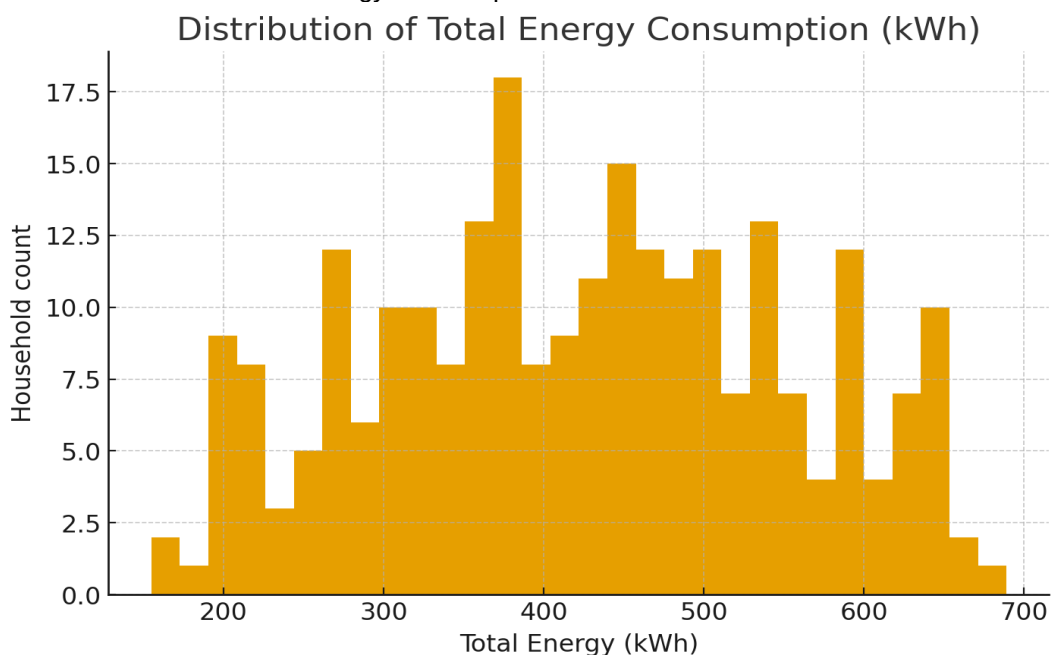
Key Findings

- Energy consumption is highly skewed – a small group of households account for disproportionately high usage.
- Correlation between energy usage and family size, income, or appliance count is weak – no single demographic predicts high consumption strongly.
- Households with many appliances or larger family sizes show only slightly higher average usage, but significant variation exists within groups.
- Seasonal variation is present (months in dataset), suggesting heating and cooling drive energy spikes.
- Top 10 households identified as extreme consumers (see detailed report).

Top 5 Recommendations

- Target top-consuming households with audits and subsidies for insulation, efficient appliances, and HVAC upgrades.
- Implement appliance efficiency rebate and trade-in programs for old refrigerators, AC units, and heaters.
- Promote behavioral change through smart meters, real-time feedback, and time-of-use pricing incentives.
- Prioritize seasonal efficiency programs (summer AC rebates, winter insulation campaigns).
- Design subsidies for low-income but high-consumption households, combining social equity with maximum energy savings.

Figure: Distribution of Total Household Energy Consumption



This analysis suggests that a focused approach — targeting outlier households and incentivizing efficiency upgrades — will provide better returns than broad demographic-based subsidies. Pilot programs should measure real-world savings before scaling citywide.